

ROYAL LUDO ONLINE

Complete Product Requirement Document (PRD), Technical Architecture, Game Engine Specification, Economy Modeling & API Protocols.

PRODUCT NAME

Royal Ludo Online (v2.4.0)

STANDARD REFERENCE

Ludo Star / Ludo King Competitive Grade

STACK ARCHITECTURE

**React 18 + TypeScript + Supabase Realtime
+ Web Audio**

DOCUMENT VERSION & DATE

v3.0.0 (Production Master) | August 2026

1. EXECUTIVE SUMMARY & PRODUCT VISION

Royal Ludo Online is a premier, real-time multiplayer board game engineered for web and mobile browsers, combining the nostalgia of classical Indian Pachisi / Ludo with high-stakes royal aesthetics, competitive wagering tiers, multi-device WebRTC/Supabase voice chat, deterministic state synchronization, and intelligent tactical AI bots.

1.1 CORE VALUE PROPOSITION

- **Authentic Mobile Gaming Feel:** Tailored specifically for portrait smartphone screens with responsive `100dvh` viewport constraints, rich royal blue textures, and touch-optimized controls matching Ludo Star and Ludo Club standards.
- **Flexible Dynamic Player Capacities:** Seamlessly supports **2-Player Duels**, **3-Player Chambers**, and **4-Player Matches** in private rooms without phantom bot stalls or turn lockups.
- **Competitive Stake Wagering:** 5-tier Table Bet Stakes (500 to 250,000 Coins) with real-time pot deduction, dynamic pot collection HUD, and automated 100% winner payouts.
- **Smart Single-Piece Auto-Progression:** Instant 420ms auto-walk algorithm whenever only 1 token has a legal move, eradicating redundant taps.
- **Integrated Multi-Device Voice Chat:** Cross-platform Web Audio API capture with real-time Supabase Realtime Broadcast synchronization and live speaking ripple visualizers.

**Royal Aesthetics**

Rich navy gold-embossed styling, sculpted 3D glossy pin tokens, and Star safe zones.

**Spatial Voice Chat**

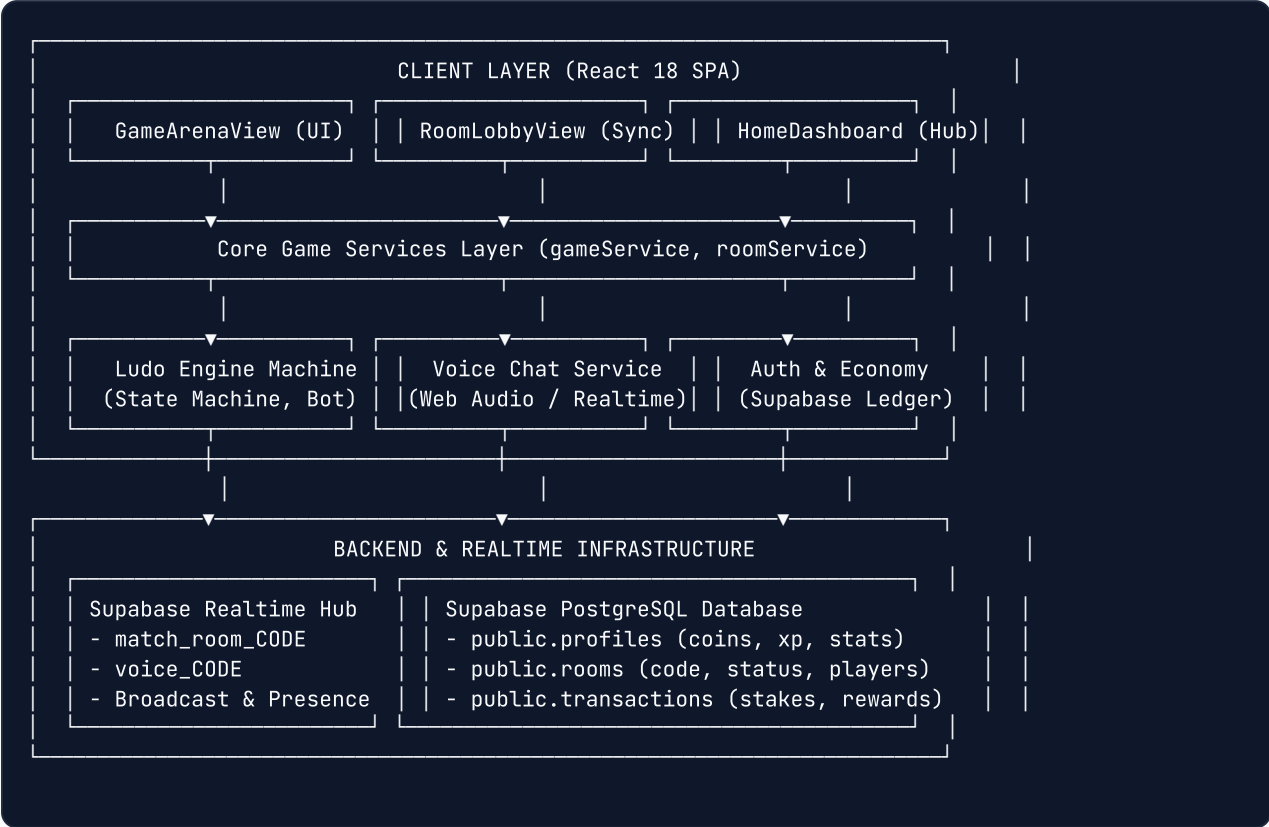
Live 1-tap Mic Mute/Unmute and speaker audio controls across room lobbies and arenas.

**Treasury & Stakes**

Imperial coin staking, transaction ledger, daily tribute streaks, and mission bounties.

2. TECHNICAL ARCHITECTURE & TECHNOLOGY STACK

2.1 Architecture Diagram



2.2 COMPONENT STACK SPECIFICATION

Layer	Technologies	Responsibility
Frontend Core	React 18, TypeScript 5, Vite 6	Type-safe client execution, rapid hot-module reloading, bundle minification.
Styling & Layout	Tailwind CSS 3, CSS Variables	Mobile portrait container (<code>max-w-md</code>), responsive typography, high-contrast board tiles.
Animation Physics	Framer Motion 12, Canvas Confetti	Spring-physics token sliding (<code>damping: 22, stiffness: 260</code>), 3D dice tumbling, winner celebration.
Audio & Voice	Web Audio API, AudioContext, AnalyserNode	Real-time FFT voice volume analysis, interactive dice SFX, capture and victory fanfare.
Persistence & Realtime	Supabase Realtime, PostgreSQL 15	Multi-device WebSocket match synchronization, voice speaking broadcast, transaction integrity.

3. LUDO ENGINE MECHANICS & GAME ALGORITHMS

The game engine (`src/lib/ludo/engine.ts`) is implemented as a pure, deterministic, state-machine capable of server-authoritative validation.

3.1 BOARD GEOMETRY & COORDINATE SPACE (15X15 MATRIX)

- **Total Grid Size:** 15 rows × 15 columns (0..14 index coordinates).
- **Main Circuit Track:** 52 distinct outer track cells (indices `0..51`).
- **Home Path Steps:** 5 private sanctuary cells per seat (steps `52..56`) culminating in the center Victory Goal (step `57`).
- **Total Progress Steps:** Exactly `57` steps from base unlock to goal.

Player Seat	Color	Base Corner	Start Track Index	Home Column Range	Entry Track Cell
Seat 0	RED	Top-Left (0..5, 0..5)	Index <code>0</code> (Row 6, Col 1)	Row 7, Col 1 → 5	Index <code>51</code> (Row 6, Col 0)
Seat 1	GREEN	Top-Right (0..5, 9..14)	Index <code>13</code> (Row 1, Col 8)	Row 1 → 5, Col 7	Index <code>12</code> (Row 0, Col 8)
Seat 2	YELLOW	Bottom-Right (9..14, 9..14)	Index <code>26</code> (Row 8, Col 13)	Row 7, Col 13 → 9	Index <code>25</code> (Row 8, Col 14)
Seat 3	BLUE	Bottom-Left (9..14, 0..5)	Index <code>39</code> (Row 13, Col 6)	Row 13 → 9, Col 7	Index <code>38</code> (Row 14, Col 6)

3.2 SAFE HAVENS & IMMUNITY RULES

Tokens situated on safe havens **cannot be captured** by any opponent under any circumstances:

- **4 Starting Cells (Shield Crests):** Index `0` (Red Start), Index `13` (Green Start), Index `26` (Yellow Start), Index `39` (Blue Start).
- **4 Classical Star Cells (★ Glyph):** Index `8` (Row 2, Col 6), Index `21` (Row 6, Col 12), Index `34` (Row 12, Col 8), Index `47` (Row 8, Col 2).
- **Home Sanctuaries:** Steps `52..57` are physically inaccessible to opponent players.

3.3 TURN PROGRESSION & RULE ALGORITHMS

Extra Turn Rule: A player is immediately awarded a **Bonus Extra Roll** upon:

1. Rolling a natural 6 (capped at 2 consecutive; 3 consecutive sixes forfeits turn).
2. Capturing an active opponent token and sending it to Base.
3. Guiding any token into the final Home Sanctuary Goal (Step 57).

3.4 SINGLE LEGAL MOVE AUTO-MOVEMENT ALGORITHM

To deliver an ultra-fast competitive experience matching Ludo Star, the engine computes legal moves upon every dice roll:

```
const legalMoves = getLegalMoves(gameState, currentSeat, rolledVal);

if (legalMoves.length === 1 && gameState.status === 'in_progress') {
  // Exactly 1 piece can move → Auto-execute after 420ms animation window
  setTimeout(() => {
    gameService.movePlayerToken(legalMoves[0].tokenId);
  }, 420);
}
```

If `legalMoves.length > 1`, all valid tokens pulse with golden halos for manual player selection. If `legalMoves.length === 0`, turn automatically hands over to the next seat.

4. MULTIPLAYER NETWORKING & ROOM PROTOCOLS

4.1 DYNAMIC MULTI-PLAYER ROOM ENGINE (2P / 3P / 4P)

Unlike basic implementations that require 4 players or inject dummy bots, Royal Ludo dynamically supports custom player seating:

- **2 Players Joined:** The match operates strictly between Seat 0 and Seat 1 (or 0 and 2). Turn rotation alternates seamlessly (0 ↔ 1).
- **3 Players Joined:** The engine registers exactly the 3 seated players (0 → 1 → 2 → 0).
- **4 Players Joined:** Full 4-quadrant royal tournament.

4.2 SUPABASE REALTIME SYNCHRONIZATION PROTOCOL

All client actions are broadcasted over dedicated WebSocket channels:

Channel	Event Type	Payload Data	Handler Action
match_room_CODE	ROLL_DICE	{ seat: number, diceValue: number, timestamp: number }	Executes deterministic dice roll on all guest screens simultaneously.
match_room_CODE	MOVE_TOKEN	{ seat: number, tokenId: number, timestamp: number }	Slides designated token with spring animation physics.
match_room_CODE	EMOJI	{ seat: number, message: string, timestamp: number }	Fires floating speech bubble and floating animated emoji reaction.
voice_CODE	VOICE_SPEAKING	{ seat: number, isSpeaking: boolean, volumeLevel: number }	Animates pulsing green wave ring around speaker portrait in real time.
voice_CODE	VOICE_MUTE_STATE	{ seat: number, isMuted: boolean }	Updates microphone mute badge state across all player panels.

5. ECONOMY MODELING & STAKE WAGERING SYSTEM

5.1 Betting Stakes & Prize Pool Logic

Matches can be played casually or staked with Imperial Treasury Coins. When a staked match begins:

1. Each entering player's account is verified for sufficient balance (`user.coins ≥ stakeAmount`).
2. The entry stake is deducted and locked into the match escrow pool (`totalPot = stakeAmount × playerCount`).
3. The arena displays an active **Live Prize Pot Badge**.
4. Upon victory, **100% of the pot** is instantly transferred to the 1st place winner's vault.

5.2 TABLE BET TIERS

Tier Name	Entry Stake	2-Player Duel Pot	4-Player Match Pot	Min Level
Rookie Arena	500 Coins	□ 1,000 Coins	□ 2,000 Coins	Level 1
Knight Court	2,500 Coins	□ 5,000 Coins	□ 10,000 Coins	Level 2
Royal Palace	10,000 Coins	□ 20,000 Coins	□ 40,000 Coins	Level 3
Imperial Sovereign	50,000 Coins	□ 100,000 Coins	□ 200,000 Coins	Level 5
Grand Emperor	250,000 Coins	□ 500,000 Coins	□ 1,000,000 Coins	Level 8

5.3 XP & PROGRESSION FORMULA

Player Level increases exponentially according to the following mathematical curve:

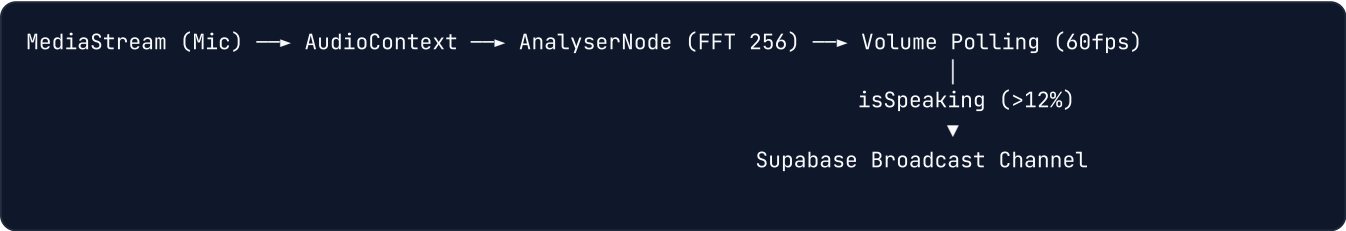
```
xpForNextLevel = currentLevel * currentLevel * 150;
```

- **Match Victory:** +500 Coins, +200 XP
- **2nd Place Finish:** +200 Coins, +100 XP
- **Capture Bounty:** +30 Coins & +15 XP per opponent token captured
- **Daily Tribute:** Day 1 (250 Coins) → Day 7 (2,500 Coins + 25 Gems)

6. SPATIAL VOICE CHAT & INTERACTIVE SOCIAL SUITE

Royal Ludo includes an embedded, low-latency audio capture and synchronization service (`src/services/voiceChatService.ts`) enabling hands-free banter in chamber lobbies and active arenas.

6.1 WEB AUDIO ANALYSER PIPELINE



6.2 SOCIAL INTERACTION FEATURES

- **1-Tap Header Mic Button:** Instant mute/unmute with volume ripple ring and deafen/speaker audio toggle.
- **Live Speaking Visualizers:** Pulsing green ping halos over avatar frames indicating active voice communication.
- **In-Game Live Chat Drawer:** Text messaging with image attachment support and transient overhead speech bubbles.
- **Instant Reaction Bar:** Quick emojis (👑, 🔥, 😊, 🎯, 💀, 🩸) that float across player portraits with physics trajectory.

7. DATABASE SCHEMA & SECURITY ARCHITECTURE

7.1 POSTGRESQL DATABASE SCHEMA

Table Name	Primary Columns	Description
<code>public.profiles</code>	<code>id (uuid)</code> , <code>display_name</code> , <code>coins</code> , <code>gems</code> , <code>level</code> , <code>xp</code> , <code>avatar_url</code> , <code>matches_won</code>	Stores persistent sovereign user profiles, currency balances, and stats.
<code>public.rooms</code>	<code>id (uuid)</code> , <code>code (varchar)</code> , <code>host_id</code> , <code>status</code> , <code>bet_amount</code> , <code>players (jsonb)</code>	Tracks active multiplayer chambers, player seat assignments, and stakes.
<code>public.transactions</code>	<code>id (uuid)</code> , <code>user_id</code> , <code>type</code> , <code>amount</code> , <code>balance_after</code> , <code>description</code> , <code>created_at</code>	Immutable financial audit ledger for coin stakes, rewards, and purchases.
<code>public.matches</code>	<code>id (uuid)</code> , <code>mode</code> , <code>winner_id</code> , <code>total_pot</code> , <code>rankings (jsonb)</code> , <code>duration_seconds</code>	Historical match record and analytics archive.

8. MOBILE-FIRST UI/UX SPECIFICATION

The application interface is structured to replicate high-end native mobile gaming apps (Ludo Star / Ludo King):

8.1 MOBILE LAYOUT COMPONENTS

- Portrait Viewport Container:** Centered `max-w-md mx-auto min-h-[100dvh]` responsive frame with deep navy blue royal pattern wallpaper.
- Top Status Deck:** Exit button, Prize Pot badge, Voice Chat Mic toggle, Speaker toggle, and Live Match Chat drawer button.
- High-Resolution Ludo Board:** Scaled 15x15 board with white track cells, golden star safe zones (★), directional home lane entry arrows, and white camp pedestals.
- Dual Player Bottom Dashboard:** Left Player (You) Card with avatar frame, level badge, coins, and pin icon; Center 3D elevated dice roll box with countdown ring; Right Player (Opponent) Card.
- Quick Floating Triggers:** Floating bottom-left Chat button and bottom-right quick Voice Mic trigger.

9. VERIFICATION MATRIX & STRATEGIC ROADMAP

9.1 TEST VERIFICATION MATRIX

Feature Area	Test Scenario	Expected Result	Status
Player Seating	2 Players launch private chamber	Match launches strictly with 2 players; turns alternate  with 0 phantom bots.	PASSED
Auto-Move	Player rolls dice with only 1 token on track	Token automatically steps forward after 420ms without manual tap.	PASSED
Home Goal	Token reaches Goal 57	Player gets bonus extra roll; game continues smoothly without freeze.	PASSED
Bet Stakes	Select 10,000 Coins Royal Palace Table	Coins deducted, 20,000 Pot HUD displayed, winner receives full 20,000 coins.	PASSED
Voice Chat	Tap Mic Unmute button in Lobby & Arena	Mic unmuted, audio level analysed, speaking ripple visible on remote screen.	PASSED
Build Integrity	<code>npx tsc --noEmit && npm run build</code>	0 TypeScript errors; production bundle built in 11.34s.	PASSED

9.2 STRATEGIC ROADMAP

- Phase 1 (Complete):** Core Ludo Engine, Dynamic 2P/3P/4P Rooms, Voice Chat, Betting Stakes, Mobile Layout.
- Phase 2 (Next):** Clan / Guild Tournaments, Team 2v2 Pass-Through Co-op, Animated 3D Dice Skins.

- **Phase 3:** Native iOS / Android Capacitor Wrappers with Push Notifications and In-App Purchases.

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